

sion. A cluster data version is surfaced in the [Interaction Model](#) when data is requested. It is used to optimize data [read](#) transactions by reducing the need to send the same data. Write interactions may also be qualified with a held data version to disallow changes, unless the cluster instance has the same data version (see [Chapter 8, Interaction Model Specification](#)). A cluster data version is published as information in some interactions (see [Chapter 8, Interaction Model Specification](#)). An externally held data version may be included as information in some interactions (see [Chapter 8, Interaction Model Specification](#)).

A cluster data version SHALL increment or be set (wrap) to zero if incrementing would exceed its maximum value. A cluster data version SHALL be maintained for each cluster instance. A cluster data version SHALL be initialized randomly when it is first published. A cluster data version SHALL be incremented if any attribute data changes.

7.10.4. New Cluster

When considering the creation of a new cluster specification, it is recommended to consider reusing and extending an existing cluster specification. These are the mechanisms to consider, in order, to extend a cluster:

1. Optional elements: attribute data, commands, events, enumerations, etc.
2. Optional feature(s) in the [FeatureMap](#) attribute for a set of elements (see 1)
3. [Cluster Aliasing](#) to reuse a cluster specification as a whole, but with a different semantic
4. [Cluster Inheritance](#)
5. A new cluster specification

7.10.5. Cluster Aliasing

Cluster aliasing allows the reuse of approved and validated specifications and derived documents, such as test plans, scripts, etc.

- More than one Cluster Identifier, each with unique purpose and semantic content, MAY map to a single cluster specification.

For example: A Concentration Measurement cluster specification may be quite abstract but have many mapped Cluster Identifiers each with a more concrete purpose, such as CO₂ or O₂ concentration measurement.

7.10.6. Cluster Inheritance

Cluster inheritance allows the reuse of approved and validated specifications and derived documents, such as test plans, scripts, etc. This allows a new cluster specification to be defined as extending or reducing the requirements of an existing cluster specification, called the base cluster. This also allows an existing cluster specification to be defined as a derived cluster, by creating a new base cluster that is more generic, allowing potential new clusters to be derived from the new base cluster.